

Active Inference BookStream #001.04 ~ "Governing Continuous Transform

BookStream | 47:34

Hello and welcome everyone. It's January 3rd, 2023. We're in active bookstream number 001.04 on governing continuous transformation. Off to you, Blue.

Great. So we are the Active Inference Institute and this discussion is over governing continuous transformation. This is a recorded and an archived live stream. So please provide us with feedback so we can improve upon our work. We are a participatory online institute that is communicating, learning and practicing applied active inference. You can reach us on social media at all of these different links here. All backgrounds and perspectives are welcome to discuss and contribute to this work and we will be following good video etiquette for live streams. If you would like to get in touch to participate in a future dot zero live stream or one of the upcoming discussions, please reach out to us via social media or email.

And again, our links. So today we are going to cover section 1.4. So we're just going to go kind of over where we are in the book, the keywords and then the different subjects and things that we've pulled out of this section. So just to start off, we will maybe just introduce ourselves. I am Blue. I am a longtime institute participant and an independent researcher in New Mexico. And I'll pass it to Daniel.

Daniel, I'm a researcher and institute participant. I'll pass to Tyler.

I'm Tyler. I'm a Dow researcher and practitioner in California.

So is there any introductory comments or do you guys just want to jump right into it?

Jump in. Okay, here we go. So this is the section that covers the free energy principle.

And that's also the title of this chapter. And I think the first keyword on our keyword slides.

So we've pulled out a lot of these keywords. I mean, we pulled them out. They weren't highlighted by Bijan or selected by him at all. And this is just kind of the topics that we thought that would be better to dive deeper into for a more thorough understanding of the material presented in this chapter. And we will, this is not like a final word. This is maybe just an introductory perspective and our take on what's maybe happening in this chapter, not a review. So we're definitely open to correction and updating our model of this model. All right. So, Tyler, do you want to read this first section for us?

Sure. So the free energy principle is the first order principle of least action empowering self-organization. FEP commands that generate the process of active inference dedicated to minimizing the information theoretical mathematical difference between top-down predictions and action generated bottom-up stimuli in pursuit of minimizing error, surprise, and entropy.

The power of Christonian active inference is fundamentally fourfold. It is a pure belief setting in dynamic and non-stationary environments. The active inference agent carries out a specific exploration to account for uncertainty by making inferences in Bayes optimal fashion. The reward signal is so characteristic of reinforcement learning is removed. And finally,

active inference sets the collaborative human-machine AI potential so integral to the multi-intelligence firm as the mathematical expression of beliefs in form of probabilities provides the very common denominator to align humans with machines.

Wow. Big if true. Okay. And the author goes on to say, there are only two distinctions that matter.

What is known and unknown and what is in our control and what is not. There are four states, active, action, internal, firm resources, sensory perception, and external, unknown and hidden behind the Markov blanket.

According to free energy governance, free energy principle-powered AI, which he uses here this capital A in lower case I to mean active inference.

So free energy principle-powered active inference is the future site of organizational becoming.

So we can unpack that, I guess, starting with this claim, which was in the abstract itself.

So maybe we don't need to reread it or was it exactly stated there? Maybe not. Oh yeah, fourfold.

So the four claims, we'll just start off this first one. The power of Prestonian actants is fundamentally fourfold.

It is a pure belief-based setting in dynamic and non-stationary environments.

So what does that mean? Like what is Prestonian active inference?

So maybe we want to just step back a little bit and talk about the FEP and the definition or explanation that Bijan picked out for this chapter is the free energy principle stems from the idea that living systems can be distinguished from other self-organizing systems

because they actively avoid deleterious phase transitions by bounding the entropy of their sensory and physical states under the FEP to be alive simply means to revisit a bounded set of states with a high probability.

And that's from this 2019 Badcock et al. paper.

And this image is from Kirchhoff et al. 2018.

And I always like this, like when I think about life and self-organizing systems, it kind of, this picture just really like speaks to what it is to kind of be self-organizing, like a hierarchical scale.

So, and this like revisiting a bounded set of states with a high probability kind of speaks to the non-equilibrium steady state density that is like a constraint of life.

We kind of have to live within, you know, like a certain temperature range, a certain, you know, glucose range,

and all life has like an optimal range of states that they stay in.

But like if your fever gets too high, like if your temperature goes too high or too low, like you will dissipate.

So you revisit this bounded set of states with a high probability and that's kind of your non-equilibrium steady state density.

Do you guys have any comments here on the FEP and self-organizing maybe?

I'll give one comment.

This is a via positiva. It's describing what the FEP is, but we could look back to those four claims and think about what this is not, the via negativa.

So rather than being belief-based in dynamic settings and adaptive autonomous agents, it might be a framework oriented around some sort of static or perennial understanding of the world.

The second point, rather than being centered around epistemic exploration and the value of reducing uncertainty and gaining information, hearing new perspectives,

it could be oriented around pragmatic or utility value, which ties very closely to the third point, which is that a framework contrary to active inference might center a reward function that maps different states of the agent or the organization or the world

to some sort of reward function, which is the heart of reward learning.

And then the fourth point also kind of ties in that rather than thinking about an ecosystem of shared intelligence,

intermediated by composable frameworks like active inference, we might be looking for like the one framework to simply rule them all,

and therefore not think about the human machine or human-human relationships as collaborative and emergent,

but rather just be looking for the one computer. We're just going to turn it on. The business is going to run. Awesome. So getting into active inference a little bit more. So we're kind of unpacking here.

We're going to unpack slowly these like four claims. And so for Stonian active inference or FEP powered active.

So in active inference, there are four states. This is what Bijan claims active action, internal, firm resources, sensory perception,

and external, and external, which is unknown and hidden behind the markup blanket. And Bijan quotes Freston here and says,

in active inference, the agent makes choices based on its belief about states of the world and not based on the value of the states.

And so that speaks to what Daniel was maybe just mentioning about reinforcement learning and we'll start to unravel reinforcement learning a little bit more.

But where reinforcement learning is really kind of driven by that pragmatic value function,

active inference is driven by your beliefs about the world. So not how much value can you gain out of moving left or moving right.

But like I fundamentally believe that I should move right because that's my policy that I've, you know, that I operate on.

So, yeah. All right. And in this, where these claims start to be unpacked in the paper,

Bijan says, one could argue that nearly all physical sciences can be reduced to a metrology in service of confirming a variant of the principle of least action. This is important because it means that physics does not offer any ground truth.

It is just search for measurements that endorse an appropriately formulated prediction based upon a variational principle.

And this quotes a Freston 2019 paper. I don't know that paper off the top of my head.

But as funny as I said, Bijan says, and then I started reading it. I'm like, no, Carl said this.

It's like a very, this sounds very much like something that would come out of a Carl paper.

Okay. And so what is this principle of least action? We've talked about this a lot before in many different live streams, but I looked it up just for like, what is like a really easy, simple way that I

could explain it in, you know, a minute or less. And I found this definition on Scholarpedia that says a true dynamical trajectory of a system between an initial and final configuration in a specified time is found by imagining all possible trajectories that the system could conceivably take computing the action, which is a functional of the trajectory. And just as a reminder, a functional is like a function of a function. So computing the action, a functional of the trajectory for each of these trajectories and selecting one that makes the action locally stationary, traditionally called least. True trajectories are those that have the least action. And so this is like the laziness, the lazy society, everybody makes the lazy choice, right? So, so like the true trajectory is the one where you kind of just stay put. Daniel, do you have a comment here?

Daniel Kahn Yes. Although on the previous slide, we looked at the particular partition where action states are the outgoing dependencies. And here we see action again with a principle of least action, but importantly to abide by a principle of least action, isn't simply to take the lowest energy action or the least movement. It's more akin to a conservation law, like the total energy is conserved of potential plus kinetic when the ball falls off the platform. So that means that there's a path of least action with the ball dropping. Now that is not the path of least movement, but it's a path of least action given an initial setup because the total energy is conserved. And so we're interested in paths of least action over the particular partitioned states, internal, external, and blanket states. And so it's a little bit like, um,

not just simply confusing, but least action can include the selection of very energetic or risky behaviors. That's cool. So, and just to kind of like, maybe tie that to a real world example, I think about like the principle of least action. Like I prefer my body temperature to be like 75 degrees all the time, right? Like, do I live in the tropics? No, because there's like a lot of things about living on a tropical island that are like not optimal for me. So like I live in New Mexico, because that's like my optimal like place, but I wear a jacket in the winter and like in the summertime, I, you know, wear less clothes or like drive around in my air conditioned car and live in my air conditioned house. So even though like, I'm still expending energy to maintain my physical temperature, like that is the path of least like action for me. Like I, I, I compensate for the change in my equilibrium state, um, through my actions. Okay. Uh, and we will talk more about energy in a minute. So unpacking this second part of the claim, the act of Asian carries out epistemic exploration to account for uncertainty by making inferences in Bayes optimal fashion. Um, okay. So like, I know what Bayes optimal is, or like, I think I know what Bayes optimal is, but, um, just to kind of like, what is Bayes optimal fashion and how do you make inference, uh, in Bayes optimal fashion? And it's cool. Cause like in a quick Google search, like I always like to search these fundamental like things, um, just to see kind of what pops up. Like, I think I know what Bayes optimal is. And, and I pulled out this 2019 paper, like what is optimal in, in optimal inference? Um, and it's interesting because like going into Bayes optimal, I mean, I think about Bayes optimal as like minimizing prediction error, or it says here, classically optimal Bayesian inference balances prior knowledge and ongoing observations to identify the model with maximum posterior probability.

Um, and so these authors actually, um, take this a step further and I, I really enjoyed their, um, like thought process here. So they said concretely, we might say that an inference procedure is optimal if it maximizes benefit per unit cost in which the benefit is some monotonically increasing function of accuracy. Um, but they proposed a generalization of this approach that, that kind of splits this cost benefit curve into two components where the first component describes the benefit, perhaps the award, the reward obtained as a function of accuracy, but then the second component, so that's the minimizing prediction error that I was speaking to earlier, right? Like in terms of accuracy. And then the second component describes accuracy as a function of costs, which can include time, memory, and computational resources needed to process information. And I think that this is like a fundamental thing that, I mean, I might overlook, like, like I might be able to get like a, a way more accurate answer. Um, like, like, you know, just so say for example, like, what is, what is the number

pi? Right. And I could just say like 3.14, I could actually go through, I could say like 3.14159, like off the top of my head, like, that's like the whatever number of digits of pi that I have memorized to like some accurate point. But like, I could come to a far more accurate answer of that if I sat there with a pencil and a piece of paper for the entire rest of my life doing long division. So I could come to like some infinite number of, of, you know, units past the decimal point. And I could go on and on and on forever. But like, is it worth it? Right? Like, is, is it worth it in terms of like memory effort to calculate that increasing accuracy? Or should is like 3.14 good enough? Right? And so that is kind of the thing that they're like the functions that they're describing here, like this benefit, this two dual component, like, yes, we want accuracy, but we also want it to be fast, or want it to be, we want to get the maximum benefit for the minimum resource expenditure. So I liked that the that they are, you know, describing accuracy as a function of the costs, the effort that goes into it. Do you guys have any comments here? Okay, so why am I going into this? Okay, so Tyler and I got into this long talk about energy. And we got into I can't remember, I guess it's not maybe in this claim, but it was somewhere in the paper, maybe I should look and try to pull it up. But we really got into discussing energy, like, what is free energy? What is like kinetic and potential energy?

And what is free energy like in FEP? And we've gone to many live streams that like dive way deeper into this. And I think it's live stream 17. That is the one I pointed you to Tyler that is Alex Kiefer does a great like unpacking unraveling of free energy in that way. But I think it's important here to point out, maybe Tyler, you can pull that quote out of the paper while I'm talking about this. But I think it's important to point out that information theoretic entropy is different than thermodynamic entropy. And so in information theory, and there are many times in the paper where, like, the author specifically says, like, we're talking about information theoretic entropy, but then it seems like that the other kind of entropy is hinted at. And so I kind of wanted to just like, draw a line between like information theoretic entropy and the other kind of entropy, like the second law of thermodynamics entropy, thermodynamic entropy. And these are these two are distinct, but not entirely unrelated. So information theory, theoretic entropy is defined as like in,

if you have a function, x , or some event x , the negative log of the probability of x is equal to the information. So here you can see like the probability versus information is this graph on the left. So where there's a high probability of an event, there's like low information, like, there's a high probability that like, I will, you know, take a breath in my in the next 30 seconds. So like me taking a breath in the next 30 seconds, like doesn't contain any information. But like the, you guys can predict that I will talk about the FEP and free energy governance in the next minute. If I start talking about like my dead dog, there's like low probability of that like happening. And so there's like maybe a much higher level of information, like, Whoa, where did this come from? This is way out of left field. So there's more information in like an unexpected claim. And so you can see like a probability distribution versus entropy that that relates like the Shannon entropy, where the entropy, like if it's if there's a high probability, it's low entropy, right? But if they're like, so so I look at this in a coin flip. So if we know that the coin is always going to come out heads, there's not any entropy in that, like, say, it's a two sided head coin. So like, it's always going to come out heads, there's no other option, the coin always flips heads. But where in a fair coin, it's 50/50, right? So like, at the right end of the second graph, it's a probability of 0.5

0.5. Like half of probability that it's heads and half of a probability that it's tails. This has the highest entropy because it's random, like, we don't know what's going to come out. And so that is what is meant here by information theoretic entropy. In contrast to thermodynamic entropy, where that is like what we think about when we think about physics and the second law of thermodynamics, entropy, we think entropy in the universe is always increasing, there's like a lot of like disorder, the disorder in the universe, like if you clean your house, and you go away for 20 years, and you come back and your house is a mess, like you're not going to be surprised at all, right? But if you, if you if your house is a mess, and you go away for 20 years, and you come back and your house is perfectly spotless, like that would be weird, right? But like, that's like, you know, scheduled for someone to come in and clean it. But but like, the end the disorder in the universe is always increasing. And that that is what is meant by entropy in terms of the second law. And we also think about this kind of entropy as energy, this is where Tyler and I got into talking about energy, but it's the entropy, the entropy is the energy that is not available to do work, right? So so in, in a system, there's like some amount of energy, like I, you know, consume so many calories in a day. And this is like very simplified, but of those calories, like I say I consumed 2000 calories in a day, I can use only like 1500 of them for work, because

some energy is always lost in heat in the conversion and any energy conversion process, like converting energy to work, you will always lose some energy as heat. And so that is kind of another way to think about this thermodynamic entropy. Tyler, were you able to find that quote where we've talked about this or no? Yeah, I was the quote is is fairly subtle, and that he goes right between information, information theory and thermodynamic theory kind of seamlessly, and he doesn't make it clear that these are like two very different, different things that there's a potential tenuous connection between. So he says FEP hence has no purpose of its own, the powers that generally the

general the process of active inference, as dedicated to minimizing the information theoretical maximal difference between the top down predictions and action generated bottom up stimuli in pursuit of minimizing errors of price and entropy. Variational free energy and thermodynamic free energy are inversely related. If variational free energy is high, then thermodynamic entropy will be high and thermodynamic free energy, the energy available for doing productive work will be low. So I think what he did there was like subtle, he like just goes between informational and thermodynamic energy energy kind of interchangeably. And they were like, wait a minute, there's like actually a difference between these two concepts and the connection between them is something that, you know, active in the the community doesn't have a great handle on at this point. Very cool. Daniel, do you have any comments there on that? It's a pretty complex and developing area how the informational entropy is related to the actual physics and biology. It's kind of like the connection between fristen free energy and gibbs free energy. How does reducing divergences on informational spaces translate to the amount of available energy and there's situations where survival could be secured one way or the other? There just isn't at this point a general answer to how they're connected. So when Tyler asked me how they're connected, what is the relationship? My answer is Maxwell's demon is the like the connection really between thermodynamic and informational free energy. And there's actually like a, I think a 2018 or 2019 paper that, that quantifies like this relationship between thermodynamic and informational entropy, um, at the quantum level. So I don't know if that applies to all scales, but it's at least cool that, and I don't remember the paper offhand, but if you search it, Maxwell's demon quantum information, you'll find it. Um, and it's pretty recent. So it's, it's neat that people are

thinking about like, like concrete and direct ways to transform, um, and translate information, theoretic and thermodynamic entropy. And so we just kind of wanted to unpack that because in that claim, it was like, Whoa, there's a lot of, a lot of things happening here that it might be more useful for the audience to have a, a deeper understanding of like what is known and what is unknown and how, how deep does

this rabbit hole really go? And right. So, so it gets pretty, uh, pretty hairy. Okay, cool. All right.

So back to reality, but back to things that we can concretely talk about. So the reward signal, so characteristic of reinforcement learning is removed and replaced with one sole purpose, surprise minimization. Effectively, the agent is empowered to minimize surprise by way of a generative model of the partially observable world, only perceiving itself and the world via outcomes. Um, and so like, just for people who might not be familiar with reinforcement learning, um, hold on just a second. Sorry. I'm getting over, getting over sickness. Sometimes I didn't want to cough it to the microphone. All right. Um, so in reinforcement learning, there's like a couple of required components. So you have a policy, which maps the state of the environment to an action. Um, and that's like for an agent, right? So this we're always talking about agentive systems here. So an agent has a policy, like if the environment is in this state, I will take this action and policies can change and update. Uh, and then there's a reward signal, which is given at like each step. So we're talking about like a time step type of thing. Second one, I take this action. My reward is two carrots, right? So

say my policy is to turn right unless I see immediate danger on my left. So, so I turn right and I turn right and I get two carrots. I turn left and like, I can't turn left because there's immediate danger over there. It's against my policy, whatever. Um, so the reward signal is the number of carrots that I receive right there at that first step. And then there's a value function, which actually tracks the reward over time. Um, and the objective in reinforcement learning is not to maximize the reward signal at each instance, but it's to maximize the value function over time. And so in a circumstance like where I turn right to get two carrots, um, but like I could have gone down and then right and gotten 50 carrots. Like, so maybe I should have gone down and right. And so this kind of, this is kind of how, uh, an agent will learn in a reinforcement learning environment to, um, maximize the value as opposed to just maximizing incrementally the reward. Um, so after every action that the agent takes, the environment sends a single number, that's the reward, the reward that the agent receives depends on the agent's action and the state. And so the agent's only goal is to maximize the total reward, which is the value function over the long term. Um, any comments here?

All right. So precision, um, Tyler, do you want to read the quote at the top of this?

Sure. So the Beijing brain has optimized not its expectations, it also has to optimize precision.

This means that one has to predict how much precision is afforded to various sources of sensory evidence relative to prior beliefs. Thank you. Um, and so this is like a definition from Wikipedia talking about, uh, like classifier systems and precision and recall in a classifier system, um, which is like a predictive system or it can be Bayesian classification. So there are, um, retrieved items like, so we're going to say, you know, of this field of images, the ones in the circle are cats, right? So we're identifying these circles, the, the, the data points within the larger circle, as all these things, the classifier says, these are all cats. The precision is of all of the things that the classifier says are cats, how many are actually cats? So it's the true positives that are in, um, in the system. And then the recall is like how many total, how many relevant items are returned. So that's, that's the, the precision here, um, that's referenced by first. Um, do you want to read this one too for us, Tyler?

Sure. All right. So when dynamics change, salience should be prevented from potentially increasing free energy in the context of corporate governance. One may think of time recognition as a one form of salience. The more senior and experienced board members are, the more likely, and they feel comfortable with mental shortcuts. So just pattern recognition built through prior experience.

Thanks. And like, when I searched for salience, I just randomly searched for like pictures of salience to make this pretty like red apple picture appear here. Um, but when I searched salience and I looked at, at images, a lot of them were like stakeholder salience. And so I like, couldn't even start to try to unpack this. Like it was some corporate definition of salience and some corporate model of salience that I don't know if, um, Bijan is referring here to more indirectly or Tyler, if you know anything about like stakeholder, the stakeholder salience model, but maybe we could ask Bijan if he, um, is referencing that, or if he's just talking about salience, the way that we talk about it as in like things that we attune our attention to because they're relevant for the task at hand or like things that stick out or things that are memorable or relevant. Right. Yeah. I don't think there's

like a corporate governance, like technical definition from it. I think it was just like you have limited attention. There are things that the portal is more likely to focus on and they have to constrain their attention on limited information. Yeah. But I was intrigued by the fact that there is like a corporate salience model. Like, Ooh, is Bijan talking about corporate salience? What's that? Um, cause it's just something not that I've never heard of before.

Yeah. Well, it's something we've seen with many active inference ontology terms, that there's a broad everyday qualitative sense, like salience is something that you pay attention to. And then there's also a more technical, usually Bayesian sense, such as salience is actually referring to the precision in terms of a variance estimator that we place on incoming sensory data. Whereas low attention means that data points coming in are not salient. They're not used to update the model. High attention is coming together with high attention to incoming data that does update the prior a lot into the posterior. And so which patterns are salient is itself learnt. And we can think about that as the regime of attention with all the strengths and weaknesses that expertise and heuristics bring along with.

Yeah. And I think, um, they, they go on to say, uh, the, the, I think the genre impacts that more in the book, um, about salience and pattern recognition.

Okay. So single and double loop learning, Tyler, do you want to take this?

Yes. Single loop learning occurs when errors are detected and corrected without altering the governing values of the master program. Double loop learning occurs when not the correct and error is necessary to alter the governing values of the master programs. Most governance models are at best single loop rule-based learning systems, but struggle to activate double loop learning.

And I found this little, like loop, this loop diagram, which is like, you know, actions and consequences, actions and consequences. And that's like the single loop, but, um, this double loop learning goes into understanding governing variables. And like, this is kind of like, I think about like meta, some kind of meta analysis, um, where you can, you know, it's zoomed out more in the double loop learning.

Yeah. With our favorite example of the temperature in the room, if the single loop is just the knob that we're controlling on the heater or the air conditioning, the double loop would be taking a step back and thinking, well, could we also open or close a window? And so if we only stayed within the single loop, maybe our heater or cooler has limited capacity. And so there's times where the environmental challenges are going to take us outside of the range of what can be learned or accommodated with a single loop. And it's those times, especially when opening up into a double loop space or like a hyper prior space is what is actually required to get the generative model operating in a way that's survivable.

So like, I would never open the window with the heater on, but boy, like I changed the weather stripping on my back door this year and it made such a huge difference. So that's like a more, a more realistic, uh, like a, a governing variable, like you have a leaky house or a poor insulation or something like that. All right. Uh, deliberate ignorance. So that is choosing not to know. Um, so Bijan talks about this and says, the more senior and experienced board members are the more likely that they feel comfortable with mental shortcuts, such as pattern recognition built through prior

experience. That's what I was going to say earlier about this. Um, again, I thought it was this next slide, but when, um, they talked about here, uh, pattern recognition as one form of salience. Um, but in this deliberate ignorance, it says, uh, it is a form of prior based heuristics or too often simply deliberate ignorance. However, in a discontinuously non-repeat game environment, pattern recognition without maintaining sensitivity for small, but nonetheless, transformative variations put survival at risk. Therefore we need a process and a measure of error that optimizes survival. Hence action centric free energy minimization, powering inactive AI. And so I thought about this, like, okay, we're talking about like corporate strategy. So it, in pattern recognition, that's super useful for something like a game of chess, right? Like, so chess has like, these are the pieces, these are the rules. There's only so many like ways that a piece can move. And it gets super complicated, but like over time, a master chess player can start to recognize patterns in the game or, or like certain, you know, sequences of moves that, that result in, uh, you know, success. And so if you know those sequences of moves that results in success, that's pattern recognition, but, but in a non-repeat game, like where the rules are always changing and the pieces can move in a different way. Like imagine now we're playing 4d chess, like where we could go also like X, Y, but we can also move in the Z direction, jump over pieces and so forth. Like if you add an extra dimension, those, um, prior patterns that you knew for, you know, XY chess now doesn't happen, right? Like they're not, they don't even apply or aren't even relevant here. So failing to update, like your, your prior pattern recognition is going to not lead to successful strategies.

Yeah. Just to make this like a little bit more concrete, like, so like a very like common example of this I've experienced is that like, sometimes when you're doing business strategy and like the business is still running and you might have customer service claims incoming while you're doing like this core business strategy. Generally, you want to ignore all of these customer service claims, cause that's just like noise and you don't want to pay attention to it. It's going to overwhelm you. Um, though occasionally you'll sometimes hear some like very unusual customer experience claims. Uh, like that's, that kind of implies something much bigger and much more serious is going on systematically within the organization. And you need to have this like spidey sense. And this is where there's often use in business settings, like the spidey sense of knowing like, Oh wait, like something bigger is going on. I need to switch gears into being super micro rather than being macro. Um, so I think that's a good example of when you have to kind of like switch your, your model of like how the organization is working really dynamically and go from deliberate ignorance, choosing what information will be selling to you.

Okay. And then sophistication, which I thought this was relevant. I can't remember the quote that we, that said this again. Oh, uh, he, Bijan actually started to unpack this in figure four. Um, and he said, we alluded to sophistication back in chapter one. Um, and this also applies to this like double loop learning. So sophisticated inference is where you have like many potential steps that can, um, unfold. And this also applies to the principle of least action, which we're talking about, which is like a decision tree search, right? Like if you're standing here at step one, you can go, you know,

which is like the top level here, you can go to any of these four points. Um, you know, you, one, you two, three, or four. So if you're here at the very beginning step, and then each one of these points has their own branch choices that you can make. And then each one of those choices has their own choices. And so this is, you know, um, the sophisticated inference, which is unpacked, um, in this paper, sophisticated inference. And then we've done on different live streams, um, here before also, we didn't do this paper, but we did deep effective inference. Um, I think that was maybe also 17, 13. I can't remember. Um, but a long time ago, it was like a 2020 early live stream, uh, with Ryan Smith. So in this sophisticated model, it's unfolding temporarily through time. And this applies to both like the trajectory, when we're talking about least action trajectories, and also to double loop learning, where you're looking at like maybe governing variables, um, over time. So, uh, he starts to unpack that here in this chapter, it goes a little bit more into it than in chapter one. All right. So now we're on the fourth part of this claim. Uh, we, we made it all the way to that. Uh, active inference should provide the basis to set free the collaborative human machine artificial intelligence potential. So integral to the multi-intelligence firm. Um, and note those two separate like AI, uh, abbreviations. All right. So Bajan makes the claim social fields are not as real time as electrically charged and chemically powered neural responsiveness, but the underlying logic and principles nonetheless determined firm performance and survival. All living self-organizing systems as is true for organizations strive to reduce entropy and pursuit of minimizing surprise. But organizations other than living systems are too often trapped in what Agurur 1995 termed skilled incompetence. Meaning that organizations are programmed to deal with error and threat in ways that are counterproductive to their own intentions. Not least because organizations struggle with unlearning and deliberately suppress subtlety. That's maybe refers back to that deliberate ignorance that we talked about, um, earlier. Do you want to talk here, Tyler, about organizational becoming? Yeah, sure. So, I mean, I just saw that last slide. I've got a couple of thoughts on that. Sure. I think about one part that was interesting about this was like, you know, explicitly connecting free energy to the firm, which I guess that's the whole point of this book, but I think that's actually quite a large claim to say that firms also exhibit free energy. And so I just want to point that out. But then also, I think in a really brilliant aspect of free energy governance is the lag between stimuli and action and action and stimuli, where that is a lot larger in an organizational space rather than something kind of like a neuron, for example. And that makes, I think, a lot harder to like operationalize and really understand free energy in like a firm context as well. So the next slide that goes into organizational becoming is this is a term he doesn't really define explicitly, um, throughout the book, but he does mention it quite a number of times. So I think it's worth calling out. So he says strategic renewal as organizational becoming is the categorical imperative of being in the game. And so again, he doesn't explicitly define this. But I think what he's implying is this idea of becoming as like this like deeper philosophical term that goes back thousands of years, where the organization isn't a static thing, or it's not something that becomes something into an end state, but rather it is like a dynamic process. And it is always a dynamic process.

And so yeah, Blue, Daniel, do you have anything you want to add to this?

You have the Heraclitus River quote, which is a statement of the dynamic nature of becoming. And then in contrast, we have being. And so people often talk about being and becoming as a noun-centric and a verb-centric way to describe something is something is that's like being, or is it becoming, which is more of a process based focus. And so he refers to it as the site of organizational becoming, not some sort of ready-made tool that's going to be, but rather a process that will be applied or is already being applied as a distributed site of becoming.

Cool. Do you want to read this one too, Tyler?

Sure. So FEP is not simply craving, but empowers a system to exploit uncertainty and complexity in pursuit of performance and survival by way of curiosity-driven epistemic forging and exploration.

Correspondingly, FENERG governance is a next-generation learning system designed for a discontinuous world, de-chaining the firm from dominant top-down logics, empowering continuous strategic renewal through purpose-directed, generative, inferential and cross-hierarchical prediction error minimization.

Nice. And so I wonder here if organizational becoming and like strategic renewal, I wonder if those are maybe being used interchangeably. We've talked about like strategic renewal or like maybe strategic renewal is like a fundamental, like an organizational becoming or organizational becoming includes strategic renewal and other things. I don't know. That's a good question for Bijan, who we will get to ask lots of questions of next week. Do you have any additional comments here, Daniel or Tyler?

Okay. Free energy governance. I'll let you take this one.

All right. So energy governance effectively transposes the active inference framework to the firm and adopts FEP as a first-order principle underlying a new logic of organizing built around three core dimensions. So structure, hierarchy, cognition, environmental enactment, capabilities, sensing, and sense-making. The framework is centered around cross-hierarchical, generative processes of sensing and sense-making to institutionalize for continuous strategic renewal as a matter of inactive inference in pursuit of firm outperformance and survival.

Thanks, Tyler. Nice. So here we are at the end of the free energy principle chapter, the end of the the first section on the FEP in the free energy governance textbook. Looking forward to having a good discussion with the author next week. Do you guys have any final thoughts on this section or chapter or anything overall?

I think a question to unpack is going to be, how does a descriptive theory of everything

in free energy principle help us make normative or even adaptive decisions for ourselves and for our organizations? Every single information architecture or strategy that a given group implements will be describable in terms of the particular partitioning into internal, external, and blanket states.

They're all going to be describable in terms of base optimal inference given some generative model and incoming stream of data. So how do we take something that is kind of like the number line? It just is and it is neutral and then proactively apply it up to and including the last mile. Do people need to actually know about information entropy to be part of a system that's designed with these strategies in mind? Or is there going to be some other way to communicate and apply it? So how we actually move it

to the space of proactive decision-making today and what that looks like in different sectors and for different organizations is really exciting.

Awesome. Well, I'm also looking forward to reading the next sections of the book and kind of going beyond like the background section into like what's next. What is the application? And looking forward to Bijan's comments next week. Tyler, any comments? Are you good? No, all good for me. Looking forward to next.

Okay. Yeah, me too. So if anybody has questions, please go ahead and submit them. You're welcome to email

them to us or submit them here in the comments, but do it before the 11th, which is when our meeting with Bijan will be. And if you want to participate on the live stream and have read the chapters in the book and watched the live streams or one or the other, get in touch. Okay. Thanks guys. Thanks.

Bye.

Thank you.